ATEASE ENGINEERING SERVICES

Submitted in partial fulfilment of the requirements for the award of

Bachelor of Engineering degree in Civil Engineering

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DEPARTMENT OF CIVIL ENGINEERING

SCHOOL OF BUILDING AND ENVIRONMENT

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY

(DEEMED TO BE UNIVERSITY)

CATEGORY -1 UNIVERSITY BY UGC

Accredited with Grade "A++" by NAAC I 12B Status by UGC I Approved by AICTE

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the Bonafide work of **AJAY.A (Reg. No.43200001), MERVIN JOSEPH (Reg No.43200005), VINOTHRAJ.V (Reg No.43200012), JEYBIN JOSE J (Reg. No.43200035)** who carried out the VENTURE CREATION Entitled "ATEASE ENGINEERING SERVICES" on June 2026 to August 2026.

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Submitted for the Viva voce Examination held on _____

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DECLARATION

We **AJAY.A , MERVIN JOSEPH, VINOTHRAJ.V, JEYBIN JOSE.J,** hereby declare that the VENTURE CREATION entitled ATEASE ENGINEERING SERVICES done by us under the guidance of Dr.B.Priyadharshini is submitted in partial fulfillment of the requirements for the award of bachelor of engineering in Civil Engineering

DATE:

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1.

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SIGNATURE OF THE CANDIDATE

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ABSTRACT

Our venture mainly focus on Building Height Lifting Using Hydraulic Jacks, presents a hydraulic lifting mechanism designed to raise heavy loads to considerable heights in a safe, controlled, and efficient manner. The system is based on Pascal's Law, which states that pressure applied to a confined fluid is transmitted equally in all directions. This principle enables hydraulic jacks to generate a large lifting force with comparatively small input effort. The proposed system consists of hydraulic jacks, a hydraulic power unit, control valves, lifting supports, a load platform, and suitable safety mechanisms.

The venture focuses on the design and fabrication of the lifting arrangement, including the selection of hydraulic components based on required load capacity and lifting height. Multiple hydraulic jacks can be arranged and synchronized to provide uniform lifting and improve system stability. The performance of the system is evaluated based on load-carrying capacity, lifting height, stability, lifting speed, and operational safety. Testing demonstrates the ability of the hydraulic system to lift heavy loads smoothly while maintaining adequate stability.

Proper structural design, synchronization, and safety provisions are essential for reliable operation, particularly when lifting loads to greater heights. The developed system can be useful in construction, building maintenance, material handling, structural lifting, and industrial applications. It provides a practical and efficient approach to vertical lifting operations and demonstrates the advantages of hydraulic power for handling heavy loads with controlled movement and reduced manual effort.

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CHAPTER 1
INTRODUCTION

1.1 Background Study

Building renovation is the process of repairing, modifying, upgrading, or improving an existing building to restore its condition, improve its performance, or adapt it to new requirements. Renovation may involve structural repairs, architectural changes, electrical and plumbing upgrades, improvement of building services, or energy-efficiency improvements. The building renovation industry developed from basic maintenance and repair activities into a specialized part of the construction industry. Earlier, renovation was mainly carried out when buildings became damaged or deteriorated. Today, renovation is also undertaken to increase the service life of buildings, improve safety, enhance appearance, reduce energy consumption, and meet modern building standards.

Common renovation activities include:

1. Repairing cracks and damaged concrete
2. Building Height lifting
3. Strengthening beams, columns and slabs
4. Repairing or replacing roofs
5. Waterproofing Re-plastering and painting
6. Replacing doors and windows
7. Upgrading electrical and plumbing systems
8. Improving ventilation
9. Modifying rooms and building layouts.

The building renovation industry is currently experiencing significant growth because of aging buildings, urbanization, changing user requirements, energy costs and sustainability objectives. Rather than replacing old buildings completely, many building owners and organizations are choosing to renovate existing structures. Renovation can be particularly attractive in locations where land is expensive or where demolition and reconstruction would cause major disruption.

1.2 Problem Identification

Rather than replacing old buildings completely, many building owners and organizations are choosing to renovate existing structures. Renovation can be particularly attractive in locations where land is expensive or where demolition and reconstruction would cause major disruption.

i. Flood protection

One of the major applications is raising buildings above flood-prone levels. Buildings that have become lower than surrounding roads or drainage levels can experience repeated waterlogging. Elevating the structure provides additional clearance between the building and ground level. This is especially relevant in areas experiencing heavy rainfall, flooding and changes in road or drainage elevations. In India, house lifting has been reported as an alternative to demolition for buildings affected by flooding and changing road levels.

ii. Foundation repair and settlement correction

Buildings may experience differential settlement because of weak soil, changes in groundwater conditions, erosion or inadequate foundations. Hydraulic jacks can be used to carefully raise or re-level portions of a structure. The lifting operation requires detailed assessment of the building's load path, foundation condition, soil properties and structural integrity before the actual lifting begins.

The problem of insufficient building height or low building elevation is mainly experienced by building owners and occupants whose buildings are located at a lower level than the surrounding ground, road, or flood level.

Homeowners in flood-prone areas People living in areas affected by frequent flooding experience water entering their houses. Repeated flooding can damage walls, flooring, furniture, electrical systems and foundations. Raising the building can provide additional clearance above the flood level. Owners of buildings affected by road-level changes When roads are repeatedly resurfaced or raised, the road level can become higher than the floor level of an existing building. This can cause rainwater to flow into the building and create drainage problems.

Owners of old or settled buildings Buildings may experience foundation settlement, causing parts of the structure to become lower than their original level. In some cases, controlled hydraulic lifting can be considered to raise or re-level the structure.

Residents in areas with poor drainage Buildings located in low-lying areas with inadequate storm-water drainage may experience waterlogging. Raising the building can reduce the risk of water entering the structure.

Commercial and industrial building owners Shops, warehouses, workshops and other commercial structures can also experience problems due to flooding, ground settlement or changes in surrounding levels. Lifting may allow the existing building to be retained instead of demolished and reconstructed.

Building owners who want to preserve an existing structure Some owners face the problem that demolition and reconstruction is expensive, time-consuming, or undesirable.

Hydraulic lifting provides a possible alternative by allowing the existing structure to be raised and a new or modified foundation to be constructed underneath.

1.3 Venture Idea

The proposed venture is to provide a building lifting and flood-protection service using hydraulic jacking technology for existing residential and small commercial buildings in Chennai. The main objective is to raise vulnerable buildings to a safer elevation without demolishing the entire structure.

The idea originated from observing the repeated flooding and waterlogging experienced by people in Chennai during heavy monsoon rainfall. Chennai is particularly vulnerable to flooding because of its low-lying areas, intense rainfall events, drainage limitations, rapid urban development and changes in natural water-flow paths. During major rainfall events, floodwater can enter houses located at low elevations. For residents, this can result in repeated expenditure on repairing floors, walls, electrical systems, furniture and other household items.

A major problem is that many existing houses cannot simply be demolished and rebuilt at a higher level, particularly when the owners have limited financial

resources or when the existing building is otherwise structurally sound. This led to the idea of using hydraulic jacking technology to raise the existing building itself. Instead of demolishing the structure, hydraulic jacks can be positioned at carefully designed lifting points and used to gradually elevate the building. Once the required height is achieved, the supporting foundation can be modified to permanently support the elevated structure.

1.4 Venture Vision

Our Venture aims to provide hydraulic building lifting services for suitable existing buildings in flood-prone areas of Chennai. The venture is motivated by the recurring problem of urban flooding and waterlogging, which causes considerable damage to residential and commercial properties.

By carefully lifting structurally suitable buildings using synchronized hydraulic jacks and providing an appropriate permanent foundation, the venture aims to increase building elevation, reduce vulnerability to floodwater and extend the useful life of existing structures. The concept provides an alternative to demolition and reconstruction and represents an innovative application of civil and hydraulic engineering for flood resilience in Chennai.

CHAPTER 2
SERVICE DESCRIPTION

2.1 Service Overview

Our Venture aims to provide Service on all renovation work and specialized in Building lifting using hydraulic jacks which is a specialized structural engineering service in which an existing building is carefully raised from its original foundation level to a higher elevation using hydraulic lifting equipment. The purpose is usually to protect buildings from flooding, waterlogging, changing road levels, ground settlement, or other site-level problems.

2.2 Costumer Pain

Customers in flood-prone areas face several problems with their existing buildings: Damage to property Floodwater can damage flooring, furniture, electrical equipment, doors, walls, and other building components. Homeowners may spend money repairing the same flood-related damage repeatedly. Moving to another property may be financially or practically difficult. Demolishing an existing building and constructing a new elevated building can require significant investment. Owners may not want to demolish a structurally sound house simply because its elevation is inadequate. Complete reconstruction can require substantial time and inconvenience. Many building owners may not know whether their existing building can be safely elevated. Electrical and plumbing systems located near the ground level can be vulnerable to floodwater. Repeated flooding can make ground-level spaces difficult to use safely.

2.3 Costumer Gain

Reduced flood exposure by increasing the building's elevation. Protection of valuable property and building components from floodwater. Potential reduction in repeated flood-repair costs. Preservation of a suitable existing building instead of automatically replacing it. Improved long-term flood resilience. Better protection of electrical and other vulnerable services when they are appropriately relocated or modified. Technology-based monitoring provides better visibility during the lifting process. Customized solution according to site conditions and structural

requirements. Reduced construction waste compared with demolition and reconstruction. Improved confidence and safety through professional structural assessment and controlled lifting. Potential increase in usable building elevation, subject to planning and structural constraints.

2.4 Service Technology

Building lifting using hydraulic jacks consists of a sequence of structural investigation, preparation, controlled lifting, and permanent support construction. The objective is to raise an existing building to a safer elevation while maintaining its structural stability.

1. Site Inspection and Feasibility Study.
2. Structural Analysis and Lifting Design.
3. Preparation of the Building Before lifting
4. Installation of Hydraulic Jacks.
5. Initial Load Transfer
6. Controlled Hydraulic Lifting
7. Continuous Monitoring
8. Construction of Permanent Supports
9. Load Transfer to Permanent Supports
10. Final Inspection and Testing
11. Reconnection and Site Restoration

Finally, electrical, water, drainage, sewage and other services are reconnected. The area around the building is restored, and appropriate drainage arrangements can be provided.



Fig 2.1: Hydraulic Jacks Used to lift building.



Fig 2.2: Building Height Rising.

CHAPTER 3

BUSINESS MODEL

3.1 Business Model Diagram

The business model connects the target customers and their needs with the proposed engineering service, key resources, activities, partners, revenue streams and costs.

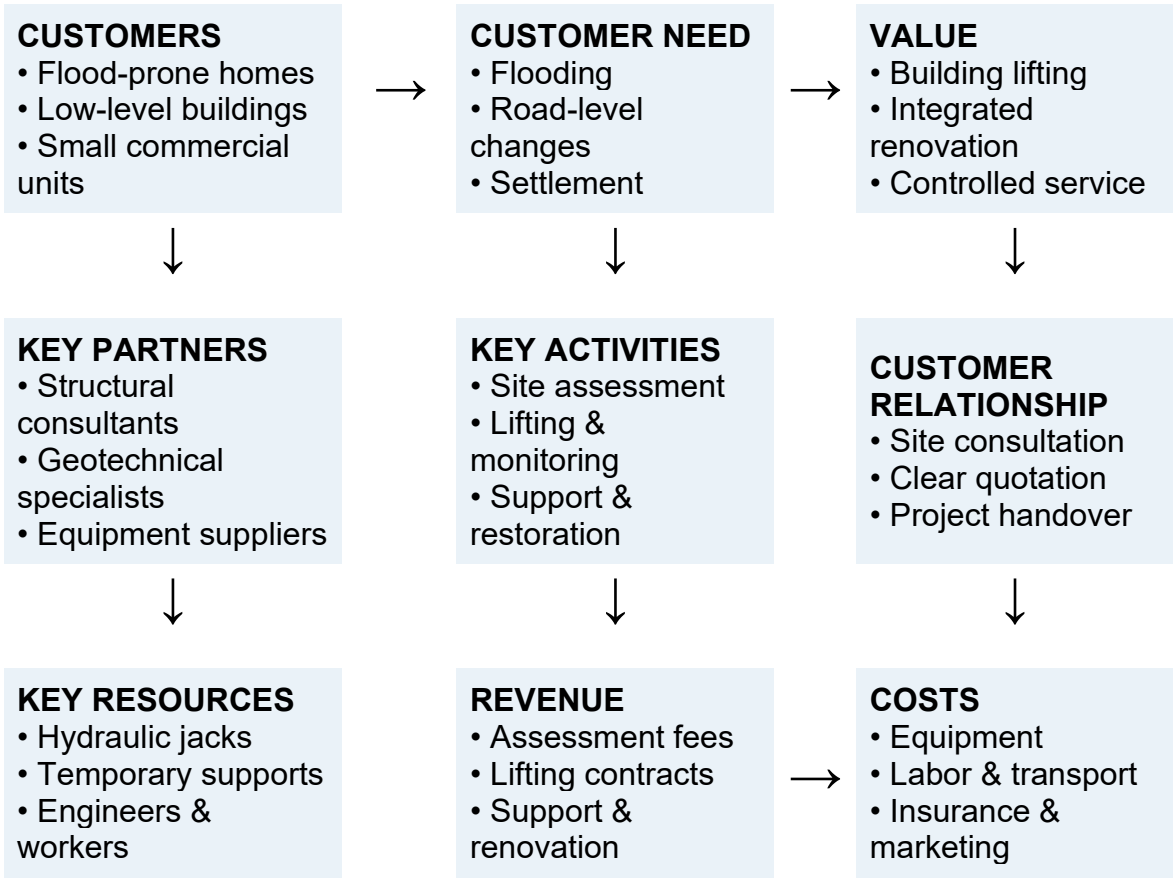


Fig 3.1: Business Model Diagram.

3.2 Lean Canvas

The Lean Canvas summarizes the venture's problem, customer segments, value proposition, solution, channels, metrics, costs, revenue streams and competitive advantage.

Table 3.1: Lean Canvas.

1. PROBLEM • Flooding / waterlogging • Low building level • Settlement • High reconstruction cost	**2. CUSTOMER SEGMENTS** • Homeowners • Residential buildings • Small commercial units • Property managers	**3. UNIQUE VALUE** Engineered, monitored building lifting for suitable existing structures.	**4. SOLUTION** • Site inspection • Structural assessment • Lifting design • Hydraulic lifting • Permanent supports	**5. CHANNELS** • Engineering referrals • Architects / contractors • Digital marketing • Direct enquiries
6. KEY METRICS • Qualified enquiries • Feasibility conversions • Projects completed • Project margin • Safety performance • Customer referrals	**7. COST STRUCTURE** • Equipment & maintenance • Labour • Transport • Temporary works • Insurance / compliance • Marketing	**8. REVENUE STREAMS** REVENUE • Assessment fees • Lifting contracts • Support works • Renovation	**9. ADVANTAGE** • Specialist civil-engineering service • Integrated lifting + renovation • Chennai flood-resilience focus	

CHAPTER 4

MARKET ANALYSIS

4.1 Market Segmentation

TAM: “Our TAM covers the broader opportunity for building elevation and renovation across Tamil Nadu.”

SAM: “Our SAM narrows this market to existing buildings in flood-prone and low-lying locations where raising the structure can be a technically relevant solution.”

SOM: “Our SOM represents the portion that a new venture can realistically target initially through limited hydraulic equipment, trained technical personnel and selected pilot projects.”

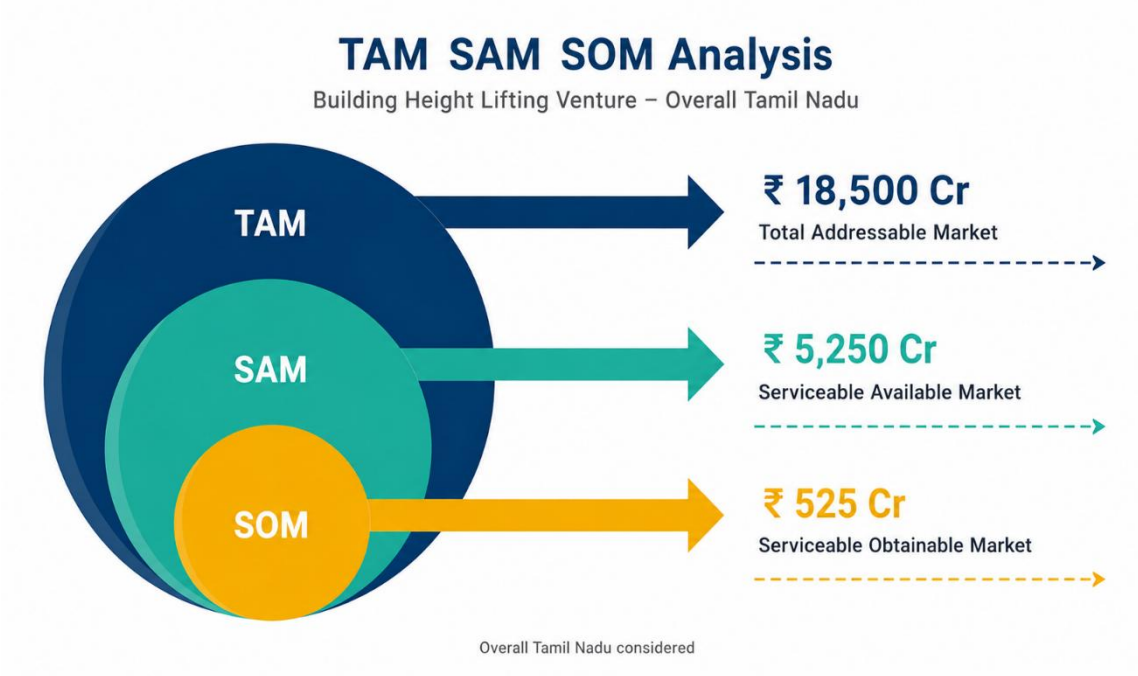


Fig 4.1: TAM,SAM,SOM of our Venture.

4.2 Customer's Survey

Questionnaire

i. What is the name of the customer?
ii. What is the age of the customer?
iii. What type of building to Renovated?
iv. When was the building Built?
v. What is plinth area of the building?
vi. Why do they consider Renovation instead of Rebuild?
vii. What was the initial basement height given?
viii. What is Height of the building to be lifted now?
ix. How Frequent they maintain their building?
x. How much is their budget and how much they expect for the service?



Fig 4.2. Customer Survey

4.3 Customer's Response

Response - 1

i. Name of the customer is Raghunanthan.
ii. Age of the customer is 61.
iii. Type of the building is Residential building G+1.
iv. The Building is built in the year 2007?
v. Plinth area of the building is 32 x 40, total 1280sqft.
vi. Renovation would match his budget at his current financial condition.
vii. The initial basement height provided was 2'6".
viii. He wants 4feet height to be elevated from ground level.
ix. He maintains his building in two ways, major and minor maintenance. Major maintenance is done 10years once and minor is done on every three years.
x. Their budget is 3 lacs and He Expect the work to completed by 4lacs.

Response - 2

i. Name of the customer is Viswananthan.
ii. Age of the customer is 46.
iii. Type of the building is Residential building G+1.
iv. The Building is built in the year 2005?
v. Plinth area of the building is 38 x 41, total 1558sqft.
vi. Renovation would match his budget at his current financial condition.
vii. The initial basement height provided was 3'.
viii. He wants 4feet height to be elevated from ground level.
ix. He maintains his building in two ways, major and minor maintenance. Major maintenance is done 10years once and minor is done on every Four years.
x. Their budget is 3 lacs and He Expect the work to completed by 3.5 lacs.



Fig 4.3. Customer Response

4.4 Competition Analysis

Table 5.1: Competition Analysis.

Benefits	**ATEASE Engineering Services**	**Mamchand House Uplifting**	**Olympic Building Lifting**	**SRBL House Lifting**
Service	Building lifting, Foundation strengthening	House lifting, building lifting & shifting	House lifting with hydraulic jacks	House lifting, structural elevation & foundation solutions
Price	Project-based quotation	Approx. ₹190+/sq. ft. listing	Project-based	Customized quotation
Technology	Synchronized hydraulic jack system	Heavy-duty hydraulic jacks	Technically advanced hydraulic jacks	Advanced/synchronized hydraulic technology
Target Customers	Old buildings, flood-prone properties, commercial buildings	Residential & commercial property owners	Residential building owners	Homeowners and property owners across India
Marketing Channel	Google, YouTube, Instagram, local contractors.	Website, online directories, direct enquiries	Website, online search	Website, online search & digital marketing

CHAPTER 5
FINANCIAL FEASIBILITY

Financial feasibility is an important part of evaluating the proposed building height lifting venture using hydraulic jacks. It determines the approximate investment required to start the business, the resources required for operation. Since this is a service-based venture, the major investment is in hydraulic lifting machinery, structural-support equipment, transportation, monitoring technology, registration, and working capital rather than in purchasing large quantities of raw materials.

5.1 Startup Cost

Table 5.1: Startup Estimation.

S. No.	Startup Cost	Estimated Cost
1	Land / Building / Office setup	₹5,00,000
2	Hydraulic lifting machinery	₹18,00,000
3	Supporting equipment	₹5,00,000
4	Technology and monitoring system	₹3,50,000
5	Registration, licenses & professional fees	₹1,00,000
6	Marketing and branding	₹1,50,000
7	Initial inventory / consumables	₹2,00,000
8	Working capital	₹8,00,000
	Total Estimated Startup Cost	**₹44,00,000**

5.2 Revenue Model

Structural Assessment Fee, Before accepting a lifting project, the company charges a separate structural feasibility assessment fee.

The assessment may include:

i. Site inspection
ii. Building condition assessment
iii. Foundation inspection
iv. Measurement of building dimensions
v. Preliminary load estimation
vi. Soil/foundation investigation

Engineering and Design, The venture charge for the professional engineering services which includes:

i. Structural analysis
ii. Hydraulic jack positioning
iii. Lifting sequence
iv. Temporary support design
v. Permanent support design
vi. Risk assessment
vii. Monitoring plan.

Equipment Rental Revenue, After the company develops sufficient equipment capacity, hydraulic jacks and monitoring equipment can potentially be rented to qualified contractors and engineering firms. This allows the company to generate revenue even when its own workforce is not carrying out a complete building-lifting project. However, because hydraulic lifting is a high-risk structural operation, equipment rental should be accompanied by appropriate technical controls, operator requirements, inspection, and contractual safeguards.

5.3 Revenue Forecast

The Revenue forecast estimates the expected number of building-lifting projects and the revenue that the proposed venture could generate during its first five years of operation. The forecast assumes that the business starts on a small scale and gradually increases its project capacity as customer awareness, technical expertise, and market acceptance improve.

Table 5.2: Revenue Estimation.

Year	Expected Projects	Average Profit/Project	Estimated Annual Sales
Year 1	12	₹80,000	₹9,60,000/-
Year 2	20	₹1.2 lakh	₹24,00,000/-
Year 3	35	₹1.2 lakh	₹42,00,000/-
Year 4	42	₹1.4 lakh	₹58,80,000/-
Year 5	60	₹1.5 lakh	₹90,00,000/-
Total			₹2,24,40,000/-(**Approx**.)

represents total projected sales of approximately ₹2.24 crore over five years.

CHAPTER 6

SUSTAINABILITY AND SOCIAL IMPACT

The venture has potential sustainability benefits because it can extend the service life of existing structures and reduce demolition where building lifting is technically feasible. The benefit is not automatic: lifting also requires steel, hydraulic equipment, transport and new support construction. Therefore, lifting should be compared with demolition and reconstruction on a project-by-project basis.

Table 6.1: Social Impact.

Area	Potential Impact	Proposed Action
Environmental benefits	Retention may reduce demolition waste	Assess reuse versus demolition before approval
Resource efficiency	Existing structural materials remain in service	Preserve suitable structural components
Waste reduction	Less demolition debris where retention is feasible	Plan selective removal and material recovery
Energy efficiency	Potentially lower embodied impact than full reconstruction	Compare alternatives during feasibility
Flood resilience	Higher level may reduce flood exposure	Coordinate elevation with drainage/flood levels
Social benefits	Reduced displacement and disruption	Plan phased works and communication
Employment	Creates specialist engineering and skilled-worker roles	Train operators, supervisors and site workers

6.1 Sustainable Development Goals

i. SDG 8 – Decent Work and Economic Growth: creates opportunities for engineers, technicians, operators and skilled workers.
ii. SDG 9 – Industry, Innovation and Infrastructure: supports an engineering-based resilience service and specialist construction capability.
iii. SDG 11 – Sustainable Cities and Communities: supports resilience of existing urban buildings where technically appropriate.
iv. SDG 12 – Responsible Consumption and Production: can reduce demolition waste and improve reuse of existing assets.
v. SDG 13 – Climate Action: contributes to adaptation/resilience against flooding and extreme rainfall.

6.2 Social Benefits

Building lifting is a high-risk construction activity. ATEASE make safety a core business requirement. No lifting should be undertaken without professional structural assessment, engineered lifting points, adequate temporary supports, an exclusion zone, competent personnel, emergency procedures and appropriate monitoring. The venture should communicate clearly that feasibility assessment may result in a building being rejected for lifting if structural or geotechnical conditions are unsuitable.

CHAPTER 7
CONCLUSION

ATEASE ENGINEERING SERVICES is a proposed specialist civil-engineering venture addressing existing buildings that are vulnerable because of low elevation, flooding, waterlogging, road-level changes or settlement. The project material identifies Chennai as the initial service area and building lifting using hydraulic jacks as the central venture idea. The proposed solution combines site inspection, feasibility assessment, structural and lifting design, hydraulic jacking, monitoring, permanent support or foundation modification and final restoration. Its main competitive advantage is integration of these activities into one service model aimed at retaining suitable existing buildings instead of automatically proceeding with demolition and reconstruction.

The market analysis indicates a potentially attractive niche among flood-vulnerable residential and small commercial properties. However, the market-size figures and illustrative customer-validation percentages are planning assumptions because actual survey responses and verified market datasets were not supplied. The next stage should focus on real customer interviews, site assessments, supplier quotations and pilot projects.

The financial model suggests that the venture can become viable if it achieves sufficient project volume, controls equipment and labour costs and maintains an adequate contribution margin. The illustrative model reaches break-even at approximately nine Year 1 projects under the stated assumptions. Actual feasibility will depend on project complexity, lifting duration, equipment ownership or rental costs, insurance, professional fees, approvals and the final price accepted by customers. Future growth can include expansion from Chennai to other flood-prone urban regions, stronger engineering and contractor partnerships, improved monitoring technology and related services such as foundation rehabilitation, drainage improvement and resilient renovation. The venture should grow in stages, with safety, engineering quality and customer trust treated as the primary conditions for expansion.